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AIM:	Setting Up a Personal Firewall Using iptables
PROBLEM STATEMENT:	Design, implement and validate a personal firewall on a Linux host using **iptables** so that the machine is protected from unauthorized incoming connections while still allowing required outgoing traffic and selected services (e.g., SSH, HTTP). Specifically: ● Start from a default (unknown) iptables state, set secure default policies (drop incoming & forwarded, allow outgoing). ● Add rules to permit only required services (example: TCP/22 for SSH, TCP/80 for HTTP). ● Add rules to explicitly block traffic from a malicious source IP. ● Make the policy persistent across reboots and verify behavior with active tests (ssh/curl/ping/nmap). This matches the experiment objective and steps in your instructions.

THEORY:	**1. Netfilter, iptables and packet flow**
	
	`iptables` is the user-level command-line tool used to manipulate the Linux kernel's **netfilter** packet-filtering framework. Netfilter intercepts network packets and processes them through a set of *tables* and *chains*:
	
	● **Tables**: collections of chains for different purposes. The most used are
	  ○ `filter` — default table for packet filtering (INPUT, OUTPUT, FORWARD chains).
	  ○ `nat` — for network address translation (not required for simple firewalling).
	  ○ `mangle` — for specialized packet alterations.
	● **Chains**:
	  ○ `INPUT` — packets destined for the local host.
	  ○ `OUTPUT` — packets generated by the host.
	  ○ `FORWARD` — packets routed through the host (when acting as a router).
	
	Packets traverse chains and are checked against rules in order until a matching rule determines the fate (ACCEPT, DROP, REJECT, etc.). If no rule matches, the chain's **policy** (default action) applies.
	
	2. Policies vs. Rules
	
	● **Policy**: default action for a chain when no rule matches (e.g., `-P INPUT DROP`).
	● **Rule**: specific match + target (e.g., allow TCP dst port 22 → `-A INPUT -p tcp --dport 22 -j ACCEPT`).
	
	A secure approach: set restrictive default policies (DROP) then explicitly add ACCEPT rules for allowed traffic. This follows the principle of *least privilege*.

	3. Stateless vs. Stateful filtering ● **Stateless rules** match individual packet fields (IP, port, protocol). **Stateful filtering** uses connection tracking (conntrack) to allow packets belonging to established/related connections without repeating per-packet rules (common usage: allow `ESTABLISHED,RELATED` on INPUT so return traffic for outgoing connections is permitted). Example rule often used: `iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT` ● Stateful handling is required for correct behavior of many protocols (TCP handshakes, FTP data connections, etc.). **4. Common rule components and flags** ● `-A INPUT` — append rule to INPUT chain. ● `-p tcp` — match TCP protocol. ● `--dport 22` — match destination port 22. ● `-s 192.168.1.100` — match source IP. ● `-j ACCEPT` / `-j DROP` — target action for matching packets. ● `iptables -L` — list current rules and policies. ● `iptables-save` / `iptables-restore` — save/restore persistent rules (distribution-specific locations: e.g., `/etc/iptables/rules.v4` on Debian/Ubuntu). Instructions- Setting Up a Pers… **5. Threat model & rationale** ● **Threats addressed**: unauthorized remote access, port scanning, targeted attacks from known malicious IPs. ● **What it doesn't solve**: kernel-level vulnerabilities, compromised privileged

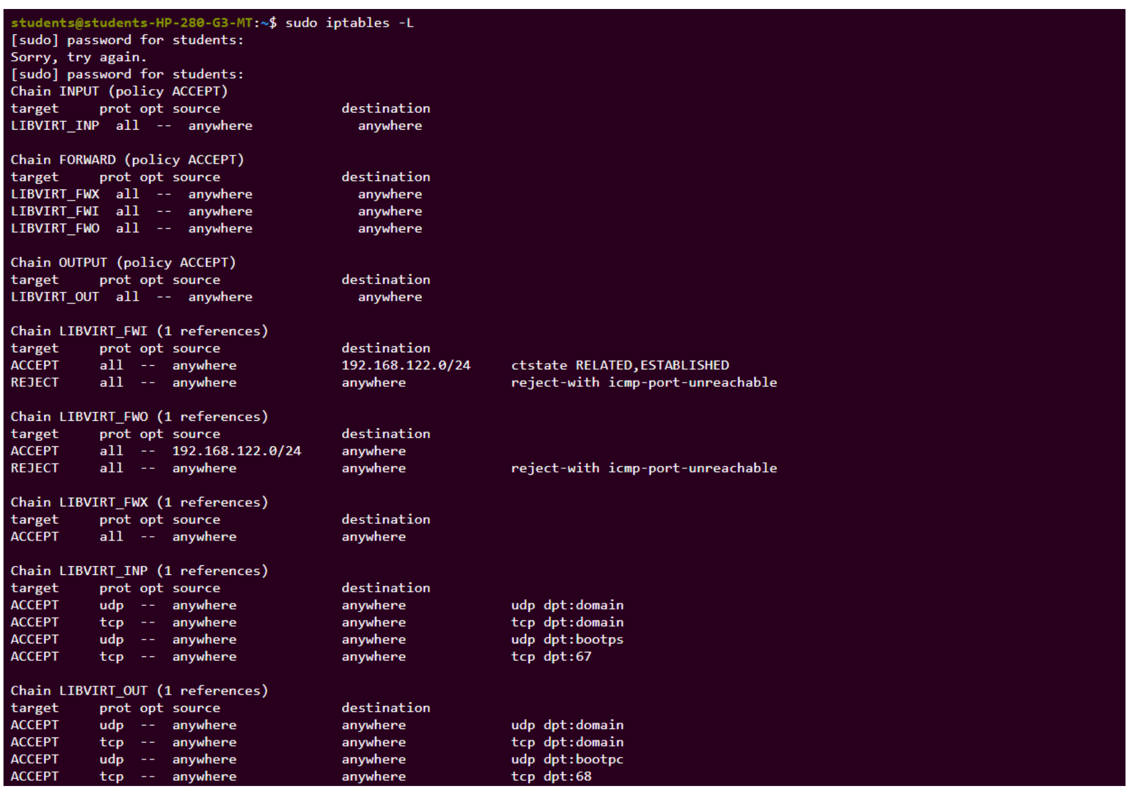
processes, application-layer bugs. Firewalling is one layer in a defense-in-depth strategy.

Implementation:

Step-by-Step Instructions for Setting Up a Personal Firewall Using iptables:

Step 1: Verify the Current iptables Configuration Check the current iptables rules:
Before making any changes, check the current configuration of iptables to see the existing rules.

sudo iptables -L



```
students@students-HP-280-G3-MT:~$ sudo iptables -L
[sudo] password for students:
Sorry, try again.
[sudo] password for students:
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
LIBVIRT_INP  all  --  anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination
LIBVIRT_FWX  all  --  anywhere             anywhere
LIBVIRT_FWI  all  --  anywhere             anywhere
LIBVIRT_FWO  all  --  anywhere             anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
LIBVIRT_OUT  all  --  anywhere             anywhere

Chain LIBVIRT_FWI (1 references)
target     prot opt source               destination
ACCEPT     all  --  anywhere             192.168.122.0/24     ctstate RELATED,ESTABLISHED
REJECT     all  --  anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWO (1 references)
target     prot opt source               destination
ACCEPT     all  --  192.168.122.0/24     anywhere
REJECT     all  --  anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWX (1 references)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere

Chain LIBVIRT_INP (1 references)
target     prot opt source               destination
ACCEPT     udp  --  anywhere             anywhere             udp dpt:domain
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:domain
ACCEPT     udp  --  anywhere             anywhere             udp dpt:bootps
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:67

Chain LIBVIRT_OUT (1 references)
target     prot opt source               destination
ACCEPT     udp  --  anywhere             anywhere             udp dpt:domain
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:domain
ACCEPT     udp  --  anywhere             anywhere             udp dpt:bootpc
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:68
```

Step 2: Set Default Policies Set default policies to block all incoming and forwarding traffic and allow outgoing traffic. This will deny all incoming connections by default while allowing the system to initiate outgoing connections.

Run the following commands:

sudo iptables -P INPUT DROP
sudo iptables -P FORWARD DROP
sudo iptables -P OUTPUT ACCEPT

- -P INPUT DROP: Blocks all incoming traffic.
- -P FORWARD DROP: Blocks all forwarded traffic.
- -P OUTPUT ACCEPT: Allows all outgoing traffic.

```
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -i lo -j ACCEPT
students@students-HP-280-G3-MT:~$ sudo iptables -A OUTPUT -o lo -j ACCEPT
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
students@students-HP-280-G3-MT:~$ sudo iptables -P INPUT DROP
students@students-HP-280-G3-MT:~$ sudo iptables -P FORWARD DROP
students@students-HP-280-G3-MT:~$ sudo iptables -P OUTPUT ACCEPT
```

Verify the updated policies:

sudo iptables -L

```
students@students-HP-280-G3-MT:~$ sudo iptables -L -v
Chain INPUT (policy DROP 2 packets, 337 bytes)
 pkts bytes target     prot opt in     out     source               destination
60289   62M LIBVIRT_INP  all  --  any    any     anywhere             anywhere
 1015 65992 ACCEPT     all  --  lo     any     anywhere             anywhere
  133 20796 ACCEPT     all  --  any    any     anywhere             anywhere             ctstate RELATED,ESTABLISHED
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:http

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 LIBVIRT_FWX  all  --  any    any     anywhere             anywhere
    0     0 LIBVIRT_FWI  all  --  any    any     anywhere             anywhere
    0     0 LIBVIRT_FWO  all  --  any    any     anywhere             anywhere

Chain OUTPUT (policy ACCEPT 8 packets, 6015 bytes)
 pkts bytes target     prot opt in     out     source               destination
30950 5284K LIBVIRT_OUT  all  --  any    any     anywhere             anywhere
 1005 65492 ACCEPT     all  --  any    lo      anywhere             anywhere

Chain LIBVIRT_FWI (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  any    virbr0  anywhere             192.168.122.0/24     ctstate RELATED,ESTABLISHED
    0     0 REJECT     all  --  any    virbr0  anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWO (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  virbr0 any     192.168.122.0/24     anywhere
    0     0 REJECT     all  --  virbr0 any     anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWX (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  virbr0 virbr0  anywhere             anywhere

Chain LIBVIRT_INP (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     udp  --  virbr0 any     anywhere             anywhere             udp dpt:domain
    0     0 ACCEPT     tcp  --  virbr0 any     anywhere             anywhere             tcp dpt:domain
    0     0 ACCEPT     udp  --  virbr0 any     anywhere             anywhere             udp dpt:bootps
```

Step 3: Allow Specific Connections Allow specific incoming traffic. For example, we want to allow incoming SSH (port 22) and HTTP (port 80) traffic while keeping the default policy to block all other incoming traffic.

Run the following commands:

sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

- -A INPUT: Appends a rule to the INPUT chain.
- -p tcp: Specifies the TCP protocol.
- --dport 22: Specifies the destination port (SSH).
- --dport 80: Specifies the destination port (HTTP).
- -j ACCEPT: Specifies that the packet should be accepted.

```
students@students-HP-280-G3-MT:~$ # Allow SSH
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
students@students-HP-280-G3-MT:~$
students@students-HP-280-G3-MT:~$ # Allow HTTP
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
students@students-HP-280-G3-MT:~$
students@students-HP-280-G3-MT:~$ # Allow related & established traffic (important for replies)
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

Verify the new rules:

sudo iptables -L

```
students@students-HP-280-G3-MT:~$ sudo iptables -L -v
Chain INPUT (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
60520   62M LIBVIRT_INP  all  --  any    any     anywhere             anywhere
 1221 76424 ACCEPT     all  --  lo     any     anywhere             anywhere
  153 23224 ACCEPT     all  --  any    any     anywhere             anywhere             ctstate RELATED,ESTABLISHED
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:http
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:http
    0     0 ACCEPT     all  --  any    any     anywhere             anywhere             ctstate RELATED,ESTABLISHED

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 LIBVIRT_FWX  all  --  any    any     anywhere             anywhere
    0     0 LIBVIRT_FWI  all  --  any    any     anywhere             anywhere
    0     0 LIBVIRT_FWO  all  --  any    any     anywhere             anywhere

Chain OUTPUT (policy ACCEPT 8 packets, 6048 bytes)
 pkts bytes target     prot opt in     out     source               destination
31174 5307K LIBVIRT_OUT  all  --  any    any     anywhere             anywhere
 1211 75924 ACCEPT     all  --  any    lo      anywhere             anywhere

Chain LIBVIRT_FWI (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  any    virbr0  anywhere             192.168.122.0/24     ctstate RELATED,ESTABLISHED
    0     0 REJECT     all  --  any    virbr0  anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWO (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  virbr0 any     192.168.122.0/24     anywhere
    0     0 REJECT     all  --  virbr0 any     anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWX (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  virbr0 virbr0  anywhere             anywhere

Chain LIBVIRT_INP (1 references)
 pkts bytes target     prot opt in     out     source               destination
```

Step 4: Block Incoming Traffic from a Specific IP Address Block incoming traffic from a specific IP address. Suppose you want to block all incoming traffic from a malicious IP (e.g., 192.168.1.100). Run the following command:

sudo iptables -A INPUT -s 192.168.1.100 -j DROP

- -A INPUT: Appends the rule to the INPUT chain.
- -s 192.168.1.100: Specifies the source IP address to block.
- -j DROP: Specifies that the packets from this IP should be dropped (blocked).

Verify the updated rules:

sudo iptables -L

```
students@students-HP-280-G3-MT:~$ sudo iptables -A INPUT -s 192.168.1.100 -j DROP
students@students-HP-280-G3-MT:~$ sudo iptables -L -v
Chain INPUT (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
60840   62M LIBVIRT_INP  all  --  any    any     anywhere             anywhere
 1497 90940 ACCEPT     all  --  lo     any     anywhere             anywhere
  196 27688 ACCEPT     all  --  any    any     anywhere             anywhere             ctstate RELATED,ESTABLISHED
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:http
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:ssh
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:http
    0     0 ACCEPT     all  --  any    any     anywhere             anywhere             ctstate RELATED,ESTABLISHED
    0     0 DROP       all  --  any    any     192.168.1.100        anywhere

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 LIBVIRT_FWX  all  --  any    any     anywhere             anywhere
    0     0 LIBVIRT_FWI  all  --  any    any     anywhere             anywhere
    0     0 LIBVIRT_FWO  all  --  any    any     anywhere             anywhere

Chain OUTPUT (policy ACCEPT 12 packets, 5944 bytes)
 pkts bytes target     prot opt in     out     source               destination
31489 5335K LIBVIRT_OUT  all  --  any    any     anywhere             anywhere
 1487 90440 ACCEPT     all  --  any    lo      anywhere             anywhere

Chain LIBVIRT_FWI (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  any    virbr0  anywhere             192.168.122.0/24     ctstate RELATED,ESTABLISHED
    0     0 REJECT     all  --  any    virbr0  anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWO (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  virbr0 any     192.168.122.0/24     anywhere
    0     0 REJECT     all  --  virbr0 any     anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWX (1 references)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  virbr0 virbr0  anywhere             anywhere
```

Step 5: Save the iptables Configuration To make the firewall rules persistent across reboots, save the iptables configuration. On Debian/Ubuntu-based systems:

sudo iptables-save > /etc/iptables/rules.v4

● This saves the current firewall rules to the specified file (/etc/iptables/rules.v4).

On CentOS/RHEL-based systems:

sudo service iptables save

```
students@students-HP-280-G3-MT:~$ sudo ls -l /etc/iptables/rules.v4
-rw-r----- 1 root root 2768 Sep 25 15:26 /etc/iptables/rules.v4
students@students-HP-280-G3-MT:~$ sudo sed -n '1,200p' /etc/iptables/rules.v4
# Generated by iptables-save v1.8.4 on Thu Sep 25 15:26:37 2025
*mangle
:PREROUTING ACCEPT [65321:62946081]
:INPUT ACCEPT [65321:62946081]
:FORWARD ACCEPT [0:0]
:OUTPUT ACCEPT [35802:5809780]
:POSTROUTING ACCEPT [35882:5818214]
:LIBVIRT_PRT - [0:0]
-A POSTROUTING -j LIBVIRT_PRT
-A LIBVIRT_PRT -o virbr0 -p udp -m udp --dport 68 -j CHECKSUM --checksum-fill
COMMIT
# Completed on Thu Sep 25 15:26:37 2025
# Generated by iptables-save v1.8.4 on Thu Sep 25 15:26:37 2025
*nat
:PREROUTING ACCEPT [173:19682]
:INPUT ACCEPT [127:15239]
:OUTPUT ACCEPT [4648:383920]
:POSTROUTING ACCEPT [4648:383920]
:LIBVIRT_PRT - [0:0]
-A POSTROUTING -j LIBVIRT_PRT
-A LIBVIRT_PRT -s 192.168.122.0/24 -d 224.0.0.0/24 -j RETURN
-A LIBVIRT_PRT -s 192.168.122.0/24 -d 255.255.255.255/32 -j RETURN
-A LIBVIRT_PRT -s 192.168.122.0/24 ! -d 192.168.122.0/24 -p tcp -j MASQUERADE --to-ports 1024-65535
-A LIBVIRT_PRT -s 192.168.122.0/24 ! -d 192.168.122.0/24 -p udp -j MASQUERADE --to-ports 1024-65535
-A LIBVIRT_PRT -s 192.168.122.0/24 ! -d 192.168.122.0/24 -j MASQUERADE
COMMIT
# Completed on Thu Sep 25 15:26:37 2025
# Generated by iptables-save v1.8.4 on Thu Sep 25 15:26:37 2025
*filter
:INPUT DROP [11:882]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [113:30179]
:LIBVIRT_FWI - [0:0]
:LIBVIRT_FWO - [0:0]
:LIBVIRT_FWX - [0:0]
```

```
:LIBVIRT_INP - [0:0]
:LIBVIRT_OUT - [0:0]
-A INPUT -i lo -j ACCEPT
-A INPUT -j LIBVIRT_INP
-A INPUT -i lo -j ACCEPT
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A INPUT -s 192.168.1.100/32 -j DROP
-A FORWARD -j LIBVIRT_FWX
-A FORWARD -j LIBVIRT_FWI
-A FORWARD -j LIBVIRT_FWO
-A OUTPUT -o lo -j ACCEPT
-A OUTPUT -j LIBVIRT_OUT
-A OUTPUT -o lo -j ACCEPT
-A LIBVIRT_FWI -d 192.168.122.0/24 -o virbr0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A LIBVIRT_FWI -o virbr0 -j REJECT --reject-with icmp-port-unreachable
-A LIBVIRT_FWO -s 192.168.122.0/24 -i virbr0 -j ACCEPT
-A LIBVIRT_FWO -i virbr0 -j REJECT --reject-with icmp-port-unreachable
-A LIBVIRT_FWX -i virbr0 -o virbr0 -j ACCEPT
-A LIBVIRT_INP -i virbr0 -p udp -m udp --dport 53 -j ACCEPT
-A LIBVIRT_INP -i virbr0 -p tcp -m tcp --dport 53 -j ACCEPT
-A LIBVIRT_INP -i virbr0 -p udp -m udp --dport 67 -j ACCEPT
-A LIBVIRT_INP -i virbr0 -p tcp -m tcp --dport 67 -j ACCEPT
-A LIBVIRT_OUT -o virbr0 -p udp -m udp --dport 53 -j ACCEPT
-A LIBVIRT_OUT -o virbr0 -p tcp -m tcp --dport 53 -j ACCEPT
-A LIBVIRT_OUT -o virbr0 -p udp -m udp --dport 68 -j ACCEPT
-A LIBVIRT_OUT -o virbr0 -p tcp -m tcp --dport 68 -j ACCEPT
COMMIT
# Completed on Thu Sep 25 15:26:37 2025
```

Verify the updated rules:

sudo iptables -L

```
students@students-HP-280-G3-MT:~$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere
LIBVIRT_INP  all  --  anywhere             anywhere
ACCEPT     all  --  anywhere             anywhere
ACCEPT     all  --  anywhere             anywhere             ctstate RELATED,ESTABLISHED
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http
ACCEPT     all  --  anywhere             anywhere             ctstate RELATED,ESTABLISHED
DROP       all  --  192.168.1.100        anywhere

Chain FORWARD (policy DROP)
target     prot opt source               destination
LIBVIRT_FWX  all  --  anywhere             anywhere
LIBVIRT_FWI  all  --  anywhere             anywhere
LIBVIRT_FWO  all  --  anywhere             anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere
LIBVIRT_OUT  all  --  anywhere             anywhere
ACCEPT     all  --  anywhere             anywhere

Chain LIBVIRT_FWI (1 references)
target     prot opt source               destination
ACCEPT     all  --  anywhere             192.168.122.0/24     ctstate RELATED,ESTABLISHED
REJECT     all  --  anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWO (1 references)
target     prot opt source               destination
ACCEPT     all  --  192.168.122.0/24     anywhere
REJECT     all  --  anywhere             anywhere             reject-with icmp-port-unreachable

Chain LIBVIRT_FWX (1 references)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere
```

Step 6: Test the Firewall Configuration by attempting to access the services you allowed and blocked:

● Try to connect to the victim machine via SSH (port 22): ssh user@target_ip

● Try to access a web server via HTTP (port 80) using a browser or a tool like curl: curl http://target_ip

● Try to ping the victim machine or connect from a blocked IP (192.168.1.100): ping target_ip

Host machine:

```
students@students-HP-280-G3-MT:~$ hostname -I        # prints space-separated IP(s) (IPv4/IPv6)
10.10.62.27 192.168.122.1
```

Machine 2:

<table>
<tr><td></td><td>

```
students@mail:~$ nmap -p 80 10.10.62.27
Starting Nmap 7.80 ( https://nmap.org ) at 2025-09-25 15:41 IST
Nmap scan report for 10.10.62.27
Host is up (0.00063s latency).

PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
students@mail:~$ curl -I http://10.10.62.27
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Thu, 25 Sep 2025 10:11:20 GMT
Content-Type: text/html
Content-Length: 10918
Last-Modified: Tue, 18 Feb 2025 08:50:21 GMT
Connection: keep-alive
ETag: "67b449cd-2aa6"
Accept-Ranges: bytes

students@mail:~$ ping -c 4 10.10.62.27
PING 10.10.62.27 (10.10.62.27) 56(84) bytes of data.
^C
--- 10.10.62.27 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3064ms

students@mail:~$ nc -vz 10.10.62.27 80
Connection to 10.10.62.27 80 port [tcp/http] succeeded!
```

</td></tr>
<tr><td>Conclusion:</td><td>We successfully configured a personal firewall using iptables to protect a Linux-based system. We set default policies to block all incoming and forwarded traffic, allowed specific connections (SSH and HTTP), and blocked traffic from a specific IP address. We also made the firewall rules persistent across reboots to ensure continuous protection.</td></tr>
</table>